

Cellular Biology: Fundamentals of Mitochondria

Mitochondria are membrane-bound cell organelles that generate most of the chemical energy needed to power the cell's metabolic processes. Chemical energy produced by the mitochondria is stored in a small molecule called adenosine triphosphate (ATP).

Structure of Mitochondria:

Mitochondria contain an outer membrane and an inner membrane. The inner membrane features numerous folds called cristae.

Glycolysis and Cellular Respiration:

Glycolysis occurs in the cytoplasm and breaks down glucose into pyruvate, yielding a net gain of 2 ATP molecules.

Pyruvate is then transported into the mitochondrial matrix where it is converted into Acetyl-CoA.

During the citric acid cycle (Krebs cycle), high-energy electrons are extracted and carried by NADH and FADH₂ to the electron transport chain.

Oxidative Phosphorylation:

The electron transport chain uses oxygen as the final electron acceptor, creating a proton gradient across the inner mitochondrial membrane.

ATP synthase utilizes this chemiosmotic proton-motive force to synthesize approximately 30 to 32 ATP molecules per glucose molecule.